

SkillSetu Sovereign National Platform

ENTERPRISE RECRUITER ROLE: ARCHITECTURE, ALGORITHMS, UX & BUSINESS MODEL CANVAS

Document Identifier:	SKILLSETU-DOC-RECRUITER-2026-V3	Classification:	Enterprise Technical Specification
Target Stakeholder:	Corporate Talent Acquisition & HR Executives	Policy Alignment:	NCrF Industry Credit Equivalence & DPDP 2023
Backend Engine:	Node.js 20 LTS + TypeScript + Express + Prisma	Inference Infrastructure:	Google Gemini 2.5 Flash + Groq LPU Llama 3 70B

1. Executive Summary & Recruiter Value Proposition

For corporate talent acquisition teams, enterprises, and startups, SkillSetu completely replaces traditional, unverified keyword-stuffed resumes with **cryptographically verified candidate competency vectors**. Recruiters can publish roles with granular NCrF-mapped skill requirements, instantly view applicant dossiers with SHA-256 block proofs, eliminate unconscious demographic bias, predict offer acceptance probabilities, and coordinate interview schedules in one unified terminal.

ZERO-BIAS MERITOCRACY ENCLAVE

Zero-Bias Blind Evaluation Protocol: Recruiters view candidate dossiers with sensitive PII (gender, caste, age, exact college name) blinded during initial screening, sorting candidates strictly by verified competency scores and validated project hashes. This guarantees merit-first hiring while achieving compliance with affirmative action benchmarks and national labor regulations.

2. Technology Stack & Language Matrix for Recruiter Services

Layer / Subsystem	Technology / Framework	Technical Rationale & Architecture Role	Concrete Recruiter Benefit
Terminal Frontend	React 18.3.1 + TypeScript + Tailwind CSS	Sovereign corporate terminal UI with high-density data tables, search filters, and status toggles.	Review dozens of candidate submissions with zero latency or browser bloat.
Navigation & Breadcrumbs	React Router DOM 6.26 + Breadcrumbs	Seamless multi-level navigation (SkillSetu > Recruiter > Applications > Candidate Review).	Rapid context switching between active vacancies and applicant pipelines.
Talent Analytics	Recharts 2.12.7 (SVG Compositor)	Applicant yield curves, hiring velocity graphs, and regional skill availability heatmaps.	Real-time executive reporting on recruitment funnel throughput and time-to-hire.
API Runtime	Node.js 20 LTS + Express 4.19 (TypeScript)	Stateless REST micro-services protected by JWT role-guard middleware (allowedRoles: ['RECRUITER']).	High-throughput applicant processing during nationwide hiring campaigns.
Data Layer	Prisma 5.17 ORM + SQLite / Turso	Relational mapping across opportunities, opportunity_skills, and applications.	Strict relational consistency; prevents lost submissions or duplicate interview bookings.
AI Predictive Engine	Google Gemini 2.5 Flash + Groq LPU Llama 3 70B	Dual-LLM inference pipeline for job fraud scanning and candidate retention prediction.	Sub-second AI assessment of applicant retention probability and offer fit.

3. Mathematical Algorithms & Predictive Intelligence Engines

Recruiter workflows are powered by deterministic scoring engines and generative predictive models:

Algorithm Name	Exact File Location	Mathematical / Technical Formulation	Recruiter Business Advantage
1. Weighted Candidate Scoring Algorithm	services/recommendation.service.ts (Lines 45-100, 242-261)	$Score = 0.70 \frac{S_{cand}}{S_{req}} \cap S_{pref} + 0.15 \frac{S_{cand}}{S_{pref}} \cap S_{prof} + 0.10 (B_{prof}) + 0.05 (B_{eval})$	Ranks applicants purely on required competencies, eliminating manual resume screening and resume spoofing.
2. AI Candidate Offer Acceptance & Retention Predictor	services/aiService.ts (Lines 246-265)	Evaluates compensation parity, geographic relocation delta, and capability surplus to output acceptance probability (0-100%) and 1-year retention index.	Prevents offer dropouts and ghosting; alerts recruiters if relocation or compensation risk requires immediate counter-action.

3. Cybercrime Job Fraud & Ghost Entity Detector	services/aiService.ts (Lines 268–285)	Natural language inspection evaluating job descriptions for illegitimate security fee demands, email domain spoofing, and unrealistic salary anomalies.	Protects platform integrity; ensures legitimate enterprises are distinguished from fraudulent employment brokers.
4. SHA-256 Cryptographic Credential Audit Engine	controllers/studentFeatures.controller.ts (Lines 50–55)	Validates candidate ledger block hashes: computes SHA-256 over candidate parameters and checks against state ledger store.	Gives recruiters 100% certainty that student certifications, CGPAs, and skills have not been forged.

4. End-to-End Recruiter User Experience (UX) Blueprint

The recruiter workflow is optimized for rapid job posting, high-precision candidate discovery, and automated interview scheduling:

Step	Interface / Route	Recruiter Action & Visual Feedback	Underlying System Telemetry
Step 1	Recruiter Authentication /login	Enterprise recruiter logs in; role-guard verifies enterprise credentials and redirects to /recruiter.	Validates organization status in users and recruiter_profiles tables.
Step 2	Enterprise Terminal /recruiter	Dashboard displays total active vacancies, applicants in pipeline, scheduled interviews, and hiring conversion rates.	Aggregated SQL queries counting application statuses and scheduled evaluations.
Step 3	Post Verified Role /recruiter/post	Form with Role Title, Location, Duration, Stipend/Salary, and Granular Required vs Preferred Skills mapped to NCrF standard ontology.	Inserts into opportunities and multiple opportunity_skills with REQUIRED/PREFERRED flags.
Step 4	Applications Review Desk /recruiter/applications	Inspect applicant roster for each posted opportunity, sorted by verified match score. Click applicant to view cryptographically verified skills and ledger proof.	Executes weighted scoring query and attaches trust ledger verification status to candidate card.
Step 5	Status Update & Action /recruiter/applications	Recruiter marks candidate as 'SHORTLISTED', 'REJECTED', or 'OFFERED'. Student receives instant real-time notification.	Updates applications.status and creates record in notifications table.
Step 6	Interview Scheduler /recruiter/interviews	Select candidate, pick evaluation date/time, add virtual meeting link (Google Meet / Zoom), and send automated calendar invites.	Persists interview record and dispatches candidate alert with calendar integration.
Step 7	Talent Analytics /recruiter/analytics	Review hiring funnel benchmarks: Application-to-Shortlist ratio, average time-to-hire, and regional talent yield.	Recharts renders funnel throughput and talent availability velocity curves.

5. API Key Architecture, Endpoints & Security Encapsulation

API Key / Parameter	Declared Location	Server Ingestion File	Operational Usage & Security Guard
GEMINI_API_KEY	backend/.env	services/aiService.ts (Lines 6–35, 246–285)	Invokes Gemini 2.5 Flash for offer retention prediction and job fraud detection. Server-only execution.
GROQ_API_KEY	backend/.env	services/aiService.ts (Lines 7, 42–69)	Ultra-fast backup engine executing Llama 3 70B via Groq LPUs for continuous candidate evaluation.
JWT_SECRET	backend/.env	middleware/auth.ts (Lines 8–18)	Guarantees enterprise session integrity; enforces role === 'RECRUITER' on all candidate review endpoints.

6. Societal & Economic Impact for Industry & Enterprises

SkillSetu modernizes corporate talent acquisition across India's economic sectors:

- **Reduction in Hiring Cycle Time:** Traditional campus recruitment takes 45–60 days per batch. SkillSetu shortens this to 48 hours through pre-verified cryptographic skill ledgers.
- **Direct Retraining Cost Savings:** Indian IT and engineering firms spend billions annually retraining graduates on foundational tools (Docker, Cloud, REST APIs). Verified NCrF credits ensure Day-1 productivity.
- **Startup & SME Talent Access:** Emerging startups gain equal visibility alongside multinational conglomerates, discovering exceptional Tier-2 talent through pure meritocratic capability matching.
- **Zero-Fraud Enterprise Hiring:** Completely eliminates candidate resume embellishment, fraudulent reference letters, and fake degree mills.

7. Business Model Canvas (BMC) — Recruiter Dimension

BMC Building Block	Enterprise Recruiter Strategic Implementation
Customer Segments	Enterprise tech corporations, Fortune 500 GCCs, high-growth startups, healthcare networks, MSMEs.
Value Propositions	Pre-verified talent with cryptographic block proofs, 90% reduction in time-to-hire, zero-bias candidate shortlisting, automated interview scheduler.
Channels	Enterprise web terminal, ATS integration webhooks (Workday, Greenhouse, Taleo), university campus relations, national job fairs.
Customer Relationships	Dedicated enterprise key account managers, quarterly talent yield reports, automated applicant dispatch notifications.
Revenue Streams	Enterprise SaaS Subscriptions: Tiered monthly/annual licenses for unlimited job posts; Pay-Per-Verified-Hire: 5-8% success fee on confirmed apprentice placement; Custom ATS API: Enterprise webhook integration.
Key Resources	National competency database, multi-dimensional weighted matching algorithm, predictive offer retention AI models.
Key Activities	Enterprise account verification, job posting fraud scanning, talent matching optimization, interview dispatch reliability.
Key Partnerships	NASSCOM, CII, FICCI, corporate hiring syndicates, college placement cells, Ministry of Skill Development (MSDE).
Cost Structure	High-throughput server infrastructure, LLM predictive inference API costs, corporate marketing, sales engineering.